

Sequences, series and Fourier analysis

Exercises Module 2

A. Find center and radius of convergence of the following series

1. $\sum_{n=0}^{\infty} 4^n (z+1)^n$
2. $\sum_{n=1}^{\infty} \frac{n^n}{n} (z - \pi i)^n$
3. $\sum_{n=0}^{\infty} \frac{n(n-1)}{3^n} (z-i)^{2n}$
4. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{4^n} (z+3)^n$
5. $\sum_{n=1}^{\infty} \frac{2^n}{n} (4z-8)^n$
6. $\sum_{n=0}^{\infty} n! (2z+1)^n$
7. $\sum_{n=1}^{\infty} \frac{(z-2i)^n}{n^n}$
8. $\sum_{n=0}^{\infty} \left(\frac{1-i}{2+3i} \right)^n z^n$
9. $\sum_{n=0}^{\infty} 16^n (z+i)^{4n}$
10. $\sum_{n=0}^{\infty} \frac{3n}{2^n n^2} z^n$

B. Calculate McLaurin series of the following functions

1. $\sin(2z^2)$

2. $\frac{1}{2+z^4}$

3. $\frac{1}{1+5iz}$

4. $\cos^2\left(\frac{z}{2}\right)$

5. $\sin^2(z)$

C. Calculate the Taylor series with center z_0 of the following functions:

1. $\frac{1}{z}, \quad z_0 = i$

2. $\frac{1}{1-z}, \quad z_0 = i$

3. $\sin(z), \quad z_0 = \frac{\pi}{2}$

4. $e^{-z}, \quad z_0 = -2$

5. $2\sin\left(\pi z + \frac{\pi}{2}\right), \quad z_0 = 0$